Point-by-point response to the reviewers of BIOINF-2026-1434

Earlier submission: *Topology Matters: The Trade-off Between Wasserstein Critics and Discriminators for Single-Cell Data Integration*, *Bioinformatics*, manuscript BIOINF-2026-1434, decision of 19 July 2026 (rejected; the editor would not consider a revised version). **This submission:** *Benchmarking alignment penalties for single-cell batch correction*, *BMC Bioinformatics*, Research article.

Rewritten on 2026-11-05 (Pass 9) against `paper/outline.tex` on branch `paper`; it replaces the response of 2026-07-22, which described experiments of an earlier design that are not in this paper. Section names below are those of the outline (Methods paragraphs by their topic; Results paragraphs Res-1 to Res-8, Res-2b and Res-5b; Discussion paragraphs Disc-1 to Disc-5). Every number below is copied from the paper text, whose numbers are listed with their source files in `paper/pass9_numbers.csv` and, from Pass 12, `paper/pass12_numbers.csv` (every abstract number; `paper/tools/check_ledger.py` re-reads the ledgers from their sources). Updated in Pass 12 (2026-11-05) after the Pass 11 rigour review. The replication results were added in Pass 14 (2026-11-06; numbers in `paper/pass14_numbers.csv`). Pass 15 (2026-11-06; `paper/pass15_changelog.md`) fixes the points of an independent re-review. Pass 17 (2026-11-06; `paper/pass17_changelog.md`) adds the finished supplement: the supplementary material of this version contains Supplementary Notes 1-12, Tables S1, S2 and S4-S11 and Figures S1-S6 (Table S3, the strength ranges of the planned fixed-strength benchmark, was removed with that benchmark). Table numbers are those the PDF prints (Pass 13): the main-text tables are numbered 1 to 3, Table 1 (methods), Table 2 (ranking) and Table 3 (placement); the strength-range table was removed.

What changed overall

We thank the three reviewers. Their comments led us to redesign the study rather than extend the earlier comparison:

1. **More penalties, one standard backbone.** Instead of one discriminator against one reference-based critic inside a custom autoencoder, the paper compares six alignment penalties added to the same stock scVI model (scvi-tools): the Jensen–Shannon discriminator, three Wasserstein-1 critics (reference, pooled and barycenter), maximum mean discrepancy (MMD) and the debiased Sinkhorn divergence, each with the decoder given the batch label (as scVI is usually run) or not (Methods, first two paragraphs; Table 1).
2. **The whole strength range, not one operating point.** Every penalty is fitted at ten strengths from 0.1 to 3,000 on eight tasks of the single-cell integration benchmark (six real, two simulated), with three seeds: 3,033

fits, none failed (Methods, "Penalty-strength sweep"; Res-1). Penalties are compared at matched batch removal, and trade-off curves, hypervolume, the share of non-dominated strengths, collapse rates and the range of good strengths describe the rest of each curve (Res-2 to Res-6).

3. **Standard tools at equal tuning effort.** Plain scVI, scvi-tools' built-in adversary, Harmony, Scanorama, sysVI, scANVI and uncorrected principal components (448 planned baseline fits, 447 fitted and scored) are compared with the penalties at equal numbers of tries (Methods, baselines paragraph; Res-7, Res-8; Tables 3 and S7; Figure 6).
4. **A replication across training seeds** refits each penalty at the two sweep strengths around the matched level with five new seeds, checked against a prediction and an agreement rule fixed before any of these fits was scored (Methods, "Replication across training seeds"; Res-2b). The rule's sign-flip test assumes the 13 task and decoder pairs are independent, although the two decoders of a task share its data and some tasks are related, so its p-value is approximate; a sensitivity analysis that flips the signs per task (both decoders together), added after the rule was fixed, is reported beside it and is approximate too, because it still flips the two tasks of a nested pair separately; a further analysis added after the rule, which flips the signs per task family (the two tasks of each nested pair and both decoders together, six families, 2^6 patterns), gives $p = 0.0156$ and is approximate too. The agreement rule is met (Res-2b; last section of this response).
5. **Direct biology readouts** per cell type (Methods, direct biology check paragraph; Res-5b; Supplementary Note 12; Table S10).
6. **Claims scoped to the evidence.** The study is described throughout as an exploratory benchmark on one backbone; the topology and reference-bottleneck explanations of the earlier paper are removed, and the limits are listed in Disc-4 and Disc-5.

Reviewer 1

R1.1 Local versus global batch removal

> "There is a mismatch between integration at local and global scales"

Changed. Local versus global removal is now one of the per-penalty analyses. The batch score is split into the three neighbourhood metrics (iLISI, kBET, graph connectivity) and the two whole-embedding metrics (principal-component regression, batch ASW), and each penalty's local-minus-global reading is taken at its crossing of the matched batch-removal level b^* (Methods, secondary-analyses paragraph; Supplementary Note 7). Res-5 and Figure 4c report it: at b^* every penalty removed more batch effect globally than locally on the ranked real tasks except the large chromatin-accessibility task, and the order among penalties changed from task to task. Disc-2 discusses it: the local-

versus-global differences do not explain the ranking. The earlier claim that the critic mixes better locally is withdrawn.

R1.2 Is there disjoint support between batches in the datasets used?

Answered in part, by the design, not by a new analysis. We did not add a measurement of support overlap. A planned check of batch separability before integration was dropped from the redesign, so the paper makes no claim that the tasks have disjoint support and no longer motivates the critic by vanishing discriminator gradients. On fairness: each penalty is now run with its standard settings (the discriminator as in scvi-tools; the critics with a gradient penalty), every penalty is compared at a batch removal that all of them reach, and every penalty's curve reached b^* on every task and decoder pair, so no penalty is scored at a level it could not attain (Methods, first paragraph and matched-batch-removal paragraph; Res-1). The Background cites the theory on aligning batches with different cell-type proportions (Zhao et al.; Wu et al.) and the sysVI study, which is the setting the reviewer describes.

R1 minor 1 Degradation with the number of batches; topological bottleneck

Removed. The claim and the bottleneck explanation are not in the paper. The number of batches is not varied (Disc-5 lists it as not tested). The redesign adds two critics that do not align to a reference batch, the pooled critic (each batch against all others) and the barycenter critic (each batch against the minibatch Wasserstein barycenter), so the reference design can be compared with joint alternatives on the same tasks (Methods, penalties paragraph; Res-2).

R1 minor 2 Cell-type annotation and marker-gene preservation

Added. Automatic annotation is tested by label transfer: each batch is held out in turn and its cell types are predicted from the other batches with a 15-nearest-neighbour classifier (Methods; Res-5; Supplementary Note 9; Supplementary Figure S3). Marker-gene preservation is part of the new direct biology check (Res-5b; Supplementary Note 12; Table S10): on the four RNA tasks the Jaccard overlap of each type's top 20 markers was 0.390 to 0.399 for every penalty and plain scVI, with differences from plain scVI of at most 0.006, whereas Harmony and PCA were lower.

R1 minor 3 Notation

Changed with the model. The custom autoencoder, and with it the cosine-similarity term, is replaced by stock scVI, whose reconstruction term is the zero-inflated negative binomial likelihood of scvi-tools. All loss equations (scVI loss, discriminator fool loss, the three critics, MMD, Sinkhorn divergence) are collected in Supplementary Note 1, where we use the notation the reviewer suggests: the likelihood written with its mean and dispersion per gene, and

the discriminator loss written as a minimisation over the classifier of the cross-entropy on each cell's posterior mean $u_i = \mu_i$, the input every penalty reads.

Reviewer 2

R2.1 The comparison is too narrow

Changed. Six penalties instead of two: one discriminator, three Wasserstein-1 critics (reference, pooled, barycenter) and two network-free distances (MMD, Sinkhorn divergence), all on one scVI backbone with two decoder settings, beside scvi-tools' own adversary and five standard tools (Methods; Table 1). The conclusions are stated for this backbone only, and Disc-4 says that the ranking may not hold for other backbones, architectures or latent sizes.

R2.2 Over-correction may be a tuning issue; show a Pareto front across the adversarial weight

Changed. No single operating point is used. Every penalty is fitted at ten strengths spanning four and a half decades, with two extra edge strengths where needed, and compared where all penalties reach the same batch removal (Methods, sweep and matched-batch-removal paragraphs). The trade-off curves (Res-3; Figure 3), the hypervolume and share of non-dominated strengths of Li et al. (Res-4; Figure 4a), the collapse rate (Res-4; Figure 4b) and the width of the range of good strengths (Res-6; Figure 5) describe each penalty over its whole range. Disc-4 states that this sweep is what Li et al. call scalarization, so it describes the penalties as users run them, not the best front a multi-objective solver could find.

R2.3 An ablation, not a mechanism; which cell types collapse, which references fail

Changed. The paper is now framed as a benchmark and the mechanistic language about topology and reference bottlenecks is removed (title, Background last paragraph, Discussion). The direct biology check now shows which cell types lose most (Res-5b): rare types lost purity in 23 of 48 task, penalty and decoder cells, and the largest drop was for the 46 ionocytes of the lung task with the batch-conditioned discriminator, whose label-transfer recall fell to 0.326 against 0.797 for plain scVI. On references: the reference batch is fixed by a rule (highest cell-type entropy, so that the target does not favour a few cell types) and is not varied (Disc-5). We report how much of each task the chosen reference covers (Methods, penalties paragraph): it holds every cell type in five of the eight tasks, and 10 of 17 (lung), 14 of 19 (human–mouse immune) and 3 of 4 (simulation 2) in the others, so in those three tasks some cells of other batches have no cell of their type in the reference. We did not vary the reference or test whether its coverage explains a penalty's performance.

R2.4 The benchmark is weak for the strength of the claims

Changed. Eight tasks instead of three (pancreas, lung, immune, human–mouse immune, small and large chromatin accessibility, two simulations), five standard tools plus plain scVI and the scvi-tools adversary, and an explicitly exploratory framing (Abstract; Background last paragraph; Methods, sweep paragraph; Disc-4). We did not add atlas-scale data (see R3.4). The replication across training seeds (last section) checks the sweep's readings on new seeds of the same tasks; it adds no new data.

R2.5 The weakness may come from the reference-based design

Agreed and tested. The pooled and barycenter critics use no reference batch (Methods, penalties paragraph). At matched batch removal the pooled critic had the best mean rank with the batch-conditioned decoder (2.20) and the reference critic 3.80, whereas with the unconditioned decoder the two were tied at 3.33 (Res-2; Table 2). The paper attributes no result to Wasserstein critics in general, and neighbouring readings are often within one standard error (Res-2; Disc-1).

R2.6 Biological over-correction is not directly validated

Added: a direct biology check (Methods; Res-5b; Supplementary Note 12; Table S10), with the four readouts the reviewer lists: per-cell-type neighbourhood purity against plain scVI, rare-cell retention (purity and label-transfer recall of types under 1% of cells), marker-gene overlap, and the label-transfer confusion table. It is exploratory: its rules were written after the sweep scores had been read and before any of its values was computed. Over the six real tasks, purity differed from plain scVI by -0.052 (batch-conditioned decoder) and -0.077 (unconditioned) for the discriminator, and by -0.019 to $+0.001$ for the reference, pooled and barycenter critics and MMD. The direct readouts agree with the summary scores in singling out the discriminator, and with the batch-conditioned decoder its loss fell more on rare types.

Reviewer 3

R3.1 Significance and generality; relation to scVI and other frameworks

Changed. The backbone is now stock scVI, the most widely used deep generative model for integration, and every penalty is compared with plain scVI and with Harmony, Scanorama, sysVI and scANVI at equal numbers of tries (Res-7, Res-8; Figure 6). With the batch-conditioned decoder, a penalty at one strength drawn at random beat plain scVI in 8 of 36 task–penalty cells and the best of six strengths in 30 of 36, a count that is optimistic because the strengths were chosen and scored on the same data; scANVI, which is trained

with the cell-type labels that several biology metrics also use, was exceeded only on pancreas (Res-8). The single backbone remains a limit, stated in Disc-4.

R3.2 Robustness to architecture and hyperparameters

Answered in part; the architecture is not varied. The relative weight of the alignment term, the hyperparameter that sets the trade-off, is swept over its whole range for every penalty (R2.2), the decoder is run with and without the batch label, and the replication across training seeds checks the readings against new seeds (Res-2b). Network depth, latent dimension (10) and hidden width (128) are fixed, as in scvi-tools' defaults. Disc-4 states this:

> "the ranking may not hold for other backbones, architectures or latent sizes"

We did not run an architecture sensitivity analysis.

R3.3 Practical guidance

Added. Disc-3 and the Conclusions now give observations from the scores on these eight tasks, not tested on independent data: the strength mattered more than the choice of penalty, because the penalties were near-tied at matched batch removal (160 of 195 within-pair differences under 0.02) while going from one to six tries raised each penalty's mean selection score by 0.012 to 0.053; plain scVI with the batch-conditioned decoder remained a strong candidate, since a penalty beat it mainly when several strengths were tried and scored on the same data (an optimistic reading); the strengths chosen on other task families lay between 0.1 and 100 across penalties with the batch-conditioned decoder; the biology was worth checking directly at the chosen strength; and scANVI scored higher than every penalty on most tasks when cell-type labels existed, although it is trained with labels that several biology metrics also use.

R3.4 Atlas-scale data

Not addressed by new data; stated as a limit. The eight tasks are of the size of the single-cell integration benchmark. Disc-4 says that the ranking may not hold for integrations of atlas scale. We did not add an atlas-scale task.

R3 minor 1 Other modalities

Answered in part. Two chromatin-accessibility tasks (mouse brain, a small and a large dataset, as gene-activity matrices) are now among the eight tasks, and their results are reported beside the RNA tasks (Res-2 to Res-6). snRNA-seq is not tested, and chromatin accessibility is analysed only as gene-activity matrices (Disc-4).

R3 minor 2 Downstream analyses

Added. Label transfer (Res-5; Supplementary Note 9), trajectory conservation inside the biology score of the two immune tasks (Methods, data-and-metrics paragraph), and the direct biology check with marker-gene overlap (Res-5b). Differential-expression testing is not added.

Replication across training seeds (relevant to R2.2, R2.4, R3.2)

Done (Pass 14, 2026-11-06). The prediction (the sweep's readings and ranks at b^* in the 13 ranked task and decoder pairs) and the agreement rule were committed under the tag `replication-prediction-v1` (commit 3a90d0f) before any replication fit was scored. The replication refits the same eight tasks with new seeds (0–4); it uses no new data. All 895 fits were scored and 0 failed: the 855 planned fits and 40 fits of the rule's extension step, which refits the next sweep strength once where the new-seed means do not bracket b^* (8 readings). After that step 77 of the 78 readings reach b^*; MMD on pancreas with the batch-conditioned decoder still does not (highest new-seed batch score 0.674 against $b^* = 0.684$) and takes the bottom place in its pair, marked as not replicating the crossing.

- Large gaps: all four sweep differences of 0.04 or more kept their sign (4 of 4). The closest to flipping was the barycenter critic minus the reference critic on the small chromatin-accessibility task with the unconditioned decoder (sweep +0.041, replication +0.013).
- Order: Kendall's τ_b between the sweep and replication ranks ranged from -0.067 to 0.867 over the 13 pairs (11 positive); the mean was 0.456, one-sided sign-flip $p = 0.00049$ over all 2^{13} sign patterns. **The agreement rule is met.**
- Before the extension step that gap could not be read (the barycenter critic had not reached b^*); the other 3 kept their sign and the mean τ_b was 0.426 ($p = 0.00049$). The rule was met only once the step it prescribes had been run; both results are reported (Res-2b).
- Sensitivity analyses, not part of the rule and defined after it was fixed: with signs flipped per task (both decoders of a task together, 2^8 patterns) $p = 0.0039$, the smallest value eight tasks allow, and still approximate, because the two tasks of a nested pair are flipped separately; with signs flipped per task family (the two tasks of each nested pair and both decoders together, 2^6 patterns over six families) $p = 0.0156$, the smallest value six families allow, also approximate; the slope-adjusted reading kept the sign of 4 of the 4 large gaps, mean τ_b 0.374, $p = 0.00073$.
- Near-ties: 155 of the 160 sweep near-ties (differences under 0.02) stayed within ± 0.04 on the new seeds. The rule tests only the sign of the four gaps of 0.04 or more (no precision calculation was made), so the replication

does not test the order of near-tied penalties; the first-ranked penalty was the same as in the sweep in 6 of 13 pairs.

Where it is reported: abstract (Results), Res-2b, Disc-1 and Supplementary Table S9; the design is in Methods ("Replication across training seeds") and Supplementary Notes 4, 10 and 11.